

Algorithms as Recipes

Project: Helpful Classroom Sorter

1. Your mission

Your learning target

I can trace an ordered algorithm using the current value after every instruction.

I can explain how changing the initial state or instruction order can change the result.

I can locate the first incorrect update and distinguish copying a value from linking two variables.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

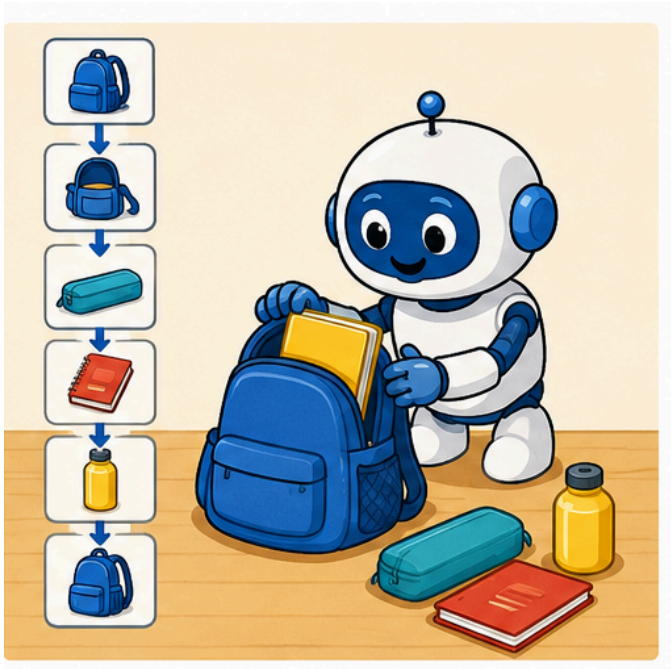
Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

An algorithm is an ordered, testable procedure, with or without code.



LEARNING TARGET On numbered squares 0–5, start at 1 facing increasing numbers. Execute FORWARD 2, TURN, FORWARD 1, STOP. Define TURN as reversing direction. Trace position and direction after every instruction. What if TURN were unspecified?

Learner name:

Readiness for the connected explanation

Two cards have the same recorded length but different widths and labels. Why can the length-only record fail to distinguish them?

This week's end goal

By the end, I can trace changing state and explain why instruction order matters.

2. Notice the evidence

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



MYSTERY A paper robot cannot infer missing intentions; it needs ordered steps that another person can execute and test.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

GOOD THINKING A prediction can change when stronger evidence appears.

A closer explanation

A classroom score display starts with the number 4. Its instructions are: add 3 to score; double score; subtract 2 from score; report score and stop. What number should it report? Before doing the arithmetic, name the value being tracked and write its starting value. Here the variable is score and the initial state is score 4.

An algorithm gives precise instructions for a task. In this chapter each instruction is carried out once, in the written order. An update replaces the current value. After adding 3, score is 7. The next instruction therefore doubles 7, not the original 4. Keeping only the starting value in mind is a common source of errors.

A variable name stays the same while its value may change. Think of score as the label on an erasable

number box. Updating the box does not keep all earlier numbers inside it. A separate trace preserves those earlier states so we can explain how the result was reached. This score procedure follows fixed arithmetic instructions; it does not learn from examples.

3. See how it works

VISUAL MECHANISM

Input → **Process** → **Output**

Follow the information instead of imagining magic inside the machine.



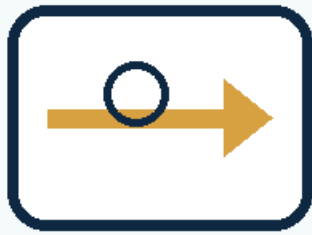
INPUT

goal, starting state, and available objects



PROCESS

follow ordered unambiguous steps



OUTPUT

completed task or a visible failure point

INPUT goal, starting state, and available objects

PROCESS follow ordered unambiguous steps

OUTPUT completed task or a visible failure point

CORE IDEA An algorithm is an ordered, testable procedure, with or without code.

UNDER THE HOOD An algorithm is a finite procedure whose steps and control flow can be executed. Correctness depends on assumptions, preconditions, order, and termination. Code is one representation of an algorithm, not the definition itself.

Draw a fresh example with all three stages:

A closer explanation

Write the initial state first. For each instruction, read the latest value, perform the operation, and record the new state. A transition is the change from one recorded state to the next.

Follow the current score through three updates

State before · Instruction · State after

State before: Score is 4

Instruction: Add 3 to score

State after: Score becomes 7

State before: Score is 7

Instruction: Double score

State after: Score becomes 14

State before: Score is 14

Instruction: Subtract 2 from score

State after: Score becomes 12

Read the rows from top to bottom. Each row starts with the state produced by the previous row. Reporting the final score gives 12; reporting does not change the value.

In words, double score means calculate twice its current value and store that result in score. Add 3 to score means increase the current value by 3. Set score to 3 means replace it with 3 instead. These instructions sound similar but do different things: if score is 7, adding 3 produces 10, while setting it to 3 produces 3.

A trace is evidence of one execution from one initial state. To compare two procedures fairly, keep the initial state the same and list every intermediate value. Then you can point to the first difference instead of only comparing the final numbers.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

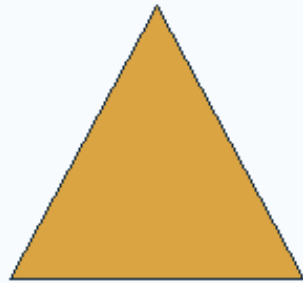
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.



Pack a bag



Make it ready



Draw a square



Choose the best book

CASE 1 — Pack a bag Clue: choose item, open bag, place item. Meaning: ordered task.

CASE 2 — Make it ready Clue: meaning is unstated. Meaning: ambiguous instruction.

CASE 3 — Draw a square Clue: four turns and moves. Meaning: testable procedure.

CASE 4 — Choose the best book Clue: needs criteria. Meaning: goal without algorithm.

BOUNDARY A goal is not yet an algorithm; each step must tell the actor what to do.

Which two pictures are easiest to confuse? What evidence separates them?

Change one thing in the pictured case

Change: Change the starting score to 0 for both recipes.

Prediction: A reports 3 and B reports 7.

Reason: A: $0 \rightarrow 0 \rightarrow 4 \rightarrow 3$. B: $0 \rightarrow 4 \rightarrow 8 \rightarrow 7$.

5. Words that unlock the idea

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





step



sequence



assumption

Word · Meaning · My example

Word: algorithm

Meaning: ordered procedure for a task

My example: _____

Word: step

Meaning: one executable action

My example: _____

Word: sequence

Meaning: the order in which steps run

My example: _____

Word: assumption

Meaning: condition taken for granted

My example: _____

Misconception detective

CLAIM TO REPAIR People will know what I meant.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word · Meaning

Word: Variable

Meaning: A named place for a value that can be updated

Word: State

Meaning: The current values being tracked at a particular point in a process

Word: Initial state

Meaning: The values before the first instruction is carried out

Word: Trace

Meaning: A record of the state as instructions are carried out in order

6. Follow a worked example

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE A paper robot cannot infer missing intentions; it needs ordered steps that another person can execute and test.

STEP 1 — OBSERVE goal, starting state, and available objects

STEP 2 — TRACE follow ordered unambiguous steps

STEP 3 — CONCLUDE completed task or a visible failure point

MODEL REASONING A strong algorithm states the starting arrangement, names each object, orders actions, and includes checks such as whether the bag is open. The revision targets the failed assumption rather than replacing every step.

DO NOT OVERCLAIM A goal is not yet an algorithm; each step must tell the actor what to do.

What new evidence could change this answer?

A closer explanation

Stage from score 4 · Add then double · Double then add

Stage from score 4: After first instruction

Add then double: Add 3 gives 7

Double then add: Double gives 8

Stage from score 4: After second instruction

Add then double: Double gives 14

Double then add: Add 3 gives 11

Stage from score 4: After subtracting 2

Add then double: Report 12

Double then add: Report 9

Both procedures use add 3, double and subtract 2, but the first two operations are exchanged. Doubling after the addition also doubles the extra 3. Adding afterward does not. The final difference comes from the order, even though both procedures started at 4.

Now use two number boxes: A starts at 2 and B starts at 0. Copy the current value of A into B, then add 5 to A. After the copy, A is 2 and B is 2. After the addition, A is 7 and B remains 2. Copying stores the value at that moment; it does not create a lasting connection between these number boxes. Record both values after each instruction, including the one that did not change.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Algorithms as Recipes

A fictional paper badge machine transforms an initial score using an ordered recipe.

The facts we will use

Start score=3. Recipe A: double, add 4, subtract 1, report.

Recipe B swaps the first two operations but keeps the same starting score.

Changing instruction order

	Start	After step 1	After step 2	Report
Recipe A	3	6	10	9
Recipe B	3	7	14	13

Read a cell using both its row and column labels.

Algorithms as Recipes

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Start

Both runs begin at 3.



2. Recipe A

3 → 6 after doubling → 10 after adding 4 → 9 after subtracting 1.



3. Recipe B

3 → 7 after adding 4 → 14 after doubling → 13 after subtracting 1.



4. Compare outputs

Same operations, different order: outputs 9 and 13.

CHANGE ONE THING

Change the starting score to 0 for both recipes.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Start

What happens and why: Both runs begin at 3.

Step: 2. Recipe A

What happens and why: 3 → 6 after doubling → 10 after adding 4 → 9 after subtracting 1.

Step: 3. Recipe B

What happens and why: $3 \rightarrow 7$ after adding 4 $\rightarrow 14$ after doubling $\rightarrow 13$ after subtracting 1.

Step: 4. Compare outputs

What happens and why: Same operations, different order: outputs 9 and 13.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Every instruction reads the current score, not the original one unless explicitly told to. In B, doubling also doubles the added 4. In A, the extra 4 is added after doubling. The final difference of 4 follows from that order, not from different starting data.

What this example does not establish

One traced input checks one run; it does not establish every recipe is equivalent or validate an entire program.

7. Find and repair a mistake

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case · Evidence · What it shows

Case: Pack a bag

Evidence: choose item, open bag, place item

What it shows: ordered task

Case: Make it ready

Evidence: meaning is unstated

What it shows: ambiguous instruction

Case: Draw a square

Evidence: four turns and moves

What it shows: testable procedure

Case: Choose the best book

Evidence: needs criteria

What it shows: goal without algorithm

Repair two tempting claims

CLAIM A People will know what I meant.

Repair:

CLAIM B An algorithm must be written in a programming language.

Repair:

STRONG ANSWER FRAME I claim _____ because the evidence shows _____. The process is _____. One limit is _____.

A closer explanation

A final value does not reveal every step that produced it. Starting at 10, adding 2 then subtracting 2 reports 10. Adding 4 then subtracting 4 also reports 10. The intermediate states differ. If a task requires a particular procedure, the final answer alone cannot establish that the required steps were followed.

Before running an algorithm, state its initial values, the meaning and order of each instruction, and when to report and stop. Our short procedures stop after their listed steps. A correct trace shows what those instructions do for the given start; it does not establish that the procedure solves every possible task or that a real-world display was measured correctly.

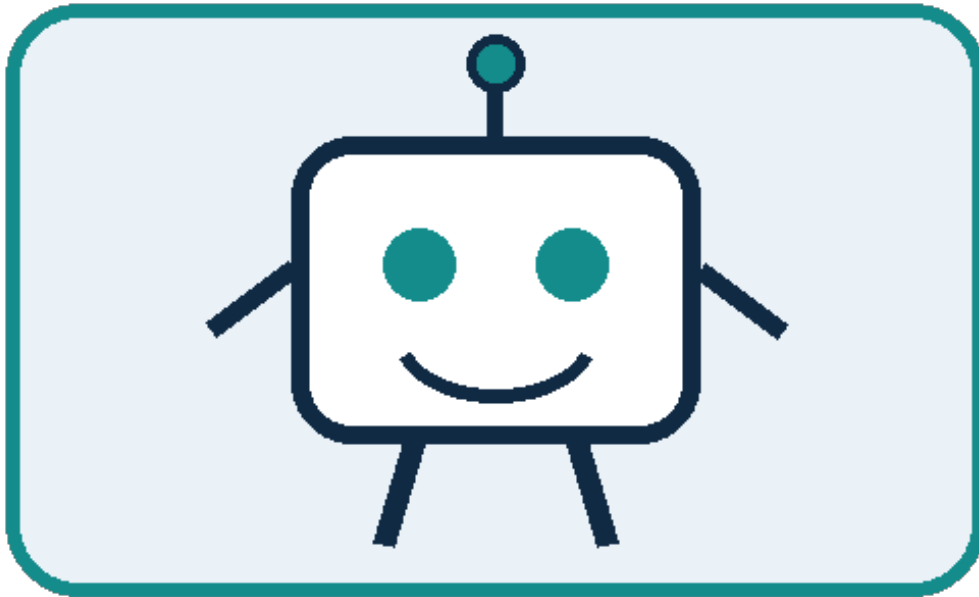
Two learners use the same instruction list but report different final scores. Name two things to compare before deciding which trace is wrong, and explain how a trace helps

8. Practise together

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: An algorithm is an ordered, testable procedure, with or without code.

Use a fresh example to explain algorithm.

Trace a new input → process → output example.

Find and repair the error in: An algorithm must be written in a programming language.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

A learner traces the score procedure as 4, 7, 8, 6. The first update is correct: 4 plus 3 is 7. The first error is the next update: double should use the current 7 and produce 14, but the learner doubled the starting 4. Correct that step and recompute the later subtraction, giving 12. Changing only the final answer would hide the faulty reasoning.

Start score at 5. Add 2 to score, double score, then subtract 1. Write the initial state and every later state. Explain which value is doubled and why

If two traces disagree, compare the initial state before blaming an instruction. If the starting values match, compare the states in order and stop at the first difference. Check the instruction used there, repair it if needed, and continue from the corrected state.

Practise with less help: Distinguish replacement from addition

Use workbook W2 for the connected example “Algorithms and changing state”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Algorithms and changing state. Proposed explanation: “Every instruction uses the original number”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Use the arrows to explain where the two zero-start runs first differ.

My explanation using a pictured step and the given facts

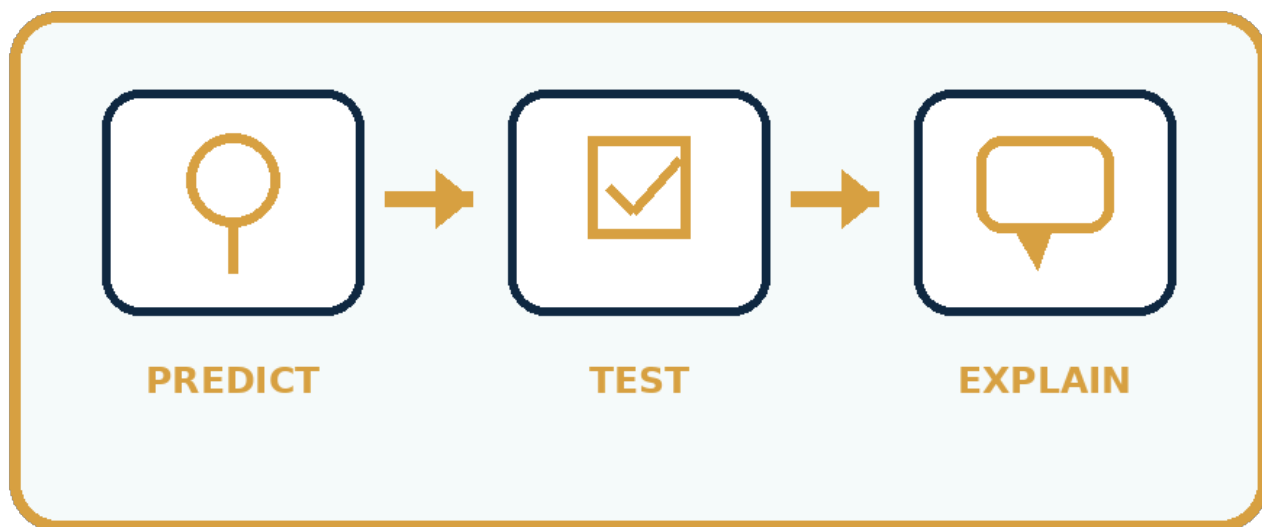
This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EXPLORER LAB

Paper Robot Bag Pack

Predict first, test honestly, and explain what the evidence means.



State your prediction before the result is known.

Complete the task: Write directions for a paper robot to pack a bag.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

Case · Prediction · Observed result · What it means

Case: Normal

Prediction: _____

Observed result: _____

What it means: _____

Case: Edge

Prediction: _____

Observed result: _____

What it means: _____

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

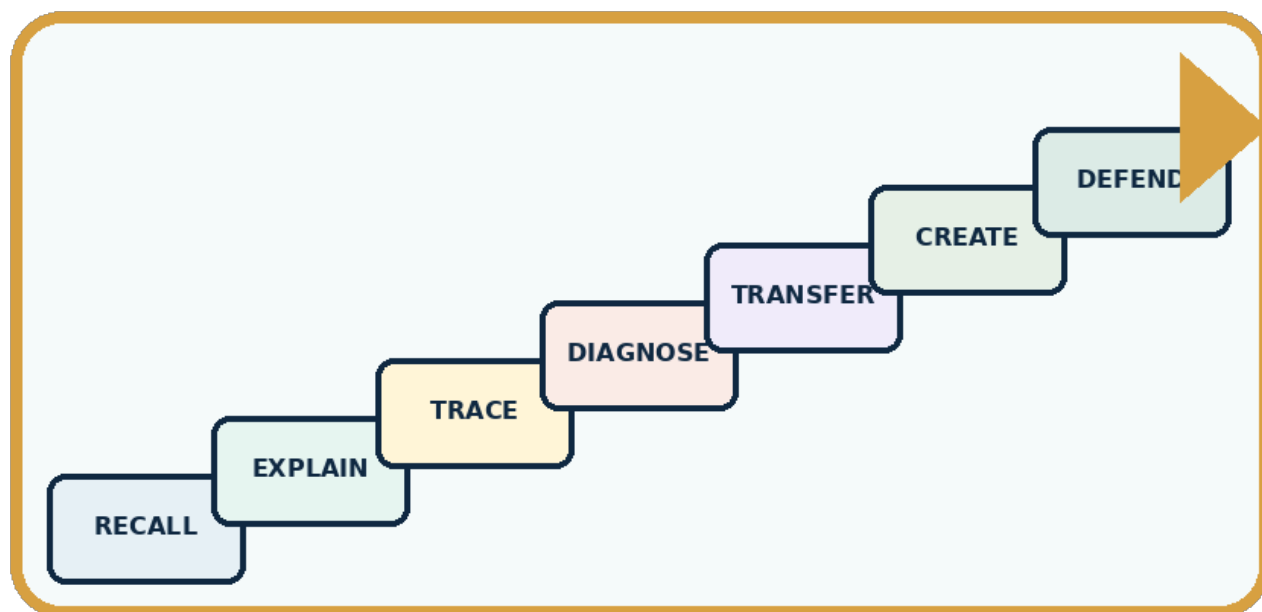
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define algorithm.

EXPLAIN An algorithm is an ordered, testable procedure, with or without code.

TRACE Show the input, process, and output.

DIAGNOSE Repair: People will know what I meant.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

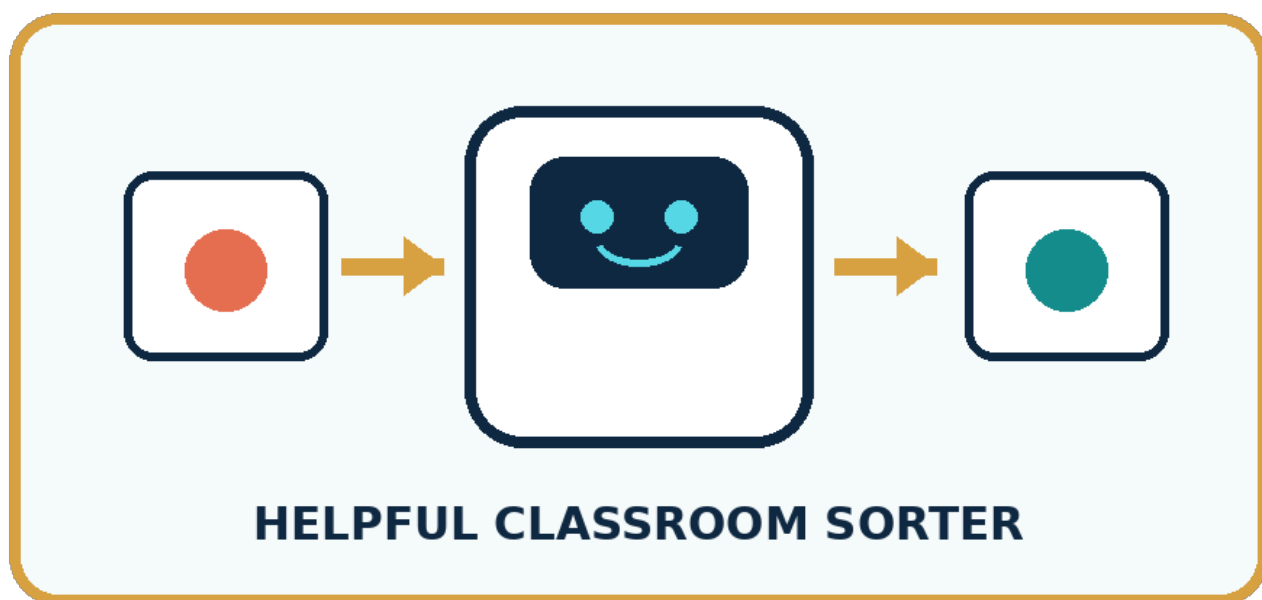
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

11. Connect your project

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



THIS WEEK'S CHECKPOINT Write the first executable sorting algorithm for one easy item before adding branches or AI.

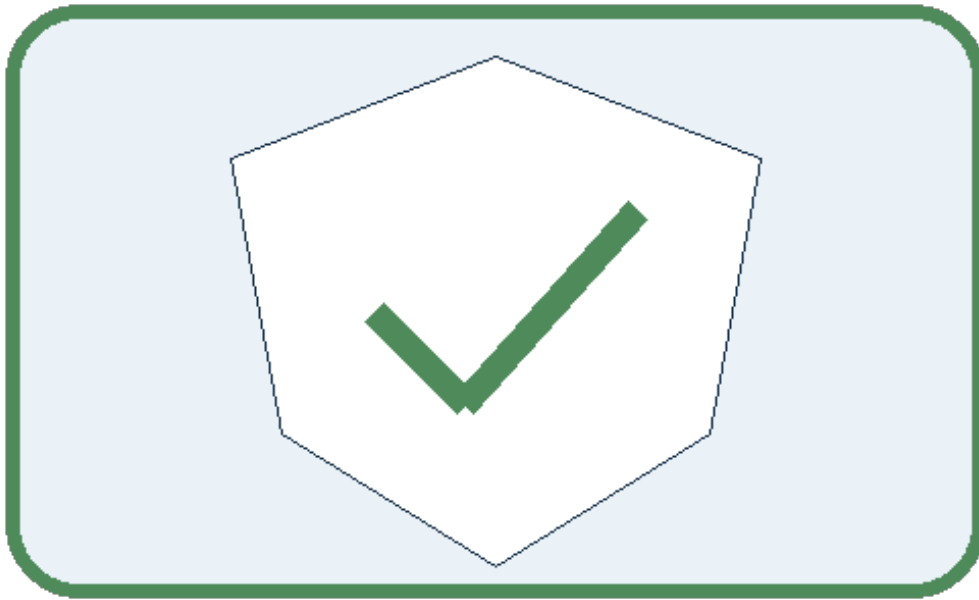
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



RESPONSIBLE DESIGN A goal is not yet an algorithm; each step must tell the actor what to do.

My project decision:

Evidence for your project

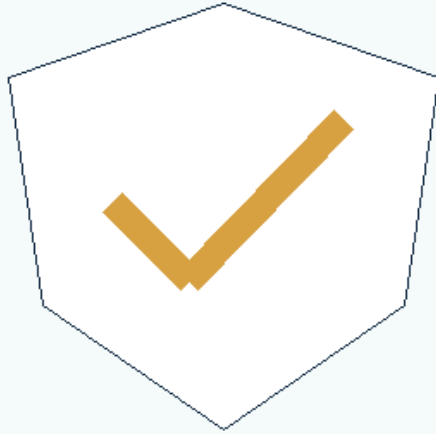
Write the sorter's steps in order, showing which values each step updates so a partner can reproduce the result.

12. Show what you know

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



INDEPENDENT PROOF

REQUIRED PROOF Write a bag-packing algorithm for a paper robot, run it exactly, and revise the first step that fails.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

SELF-CHECK A strong algorithm states the starting arrangement, names each object, orders actions, and includes checks such as whether the bag is open. The revision targets the failed assumption rather than replacing every step.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

NEXT BRIDGE Carry today's evidence habit into: If/Then Decisions.

13. Study a complete case

Week 9 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This

case extends the illustrated lesson.

The situation

A robot starts on the left of three squares, facing right. It executes MOVE, MOVE, STOP, each MOVE being one square.

The question

Trace the position after each command.

Worked explanation

Middle, rightmost, unchanged. Direction, distance and starting conditions must be specified.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Trace a robot's position

Evidence or stage · Value or result

Evidence or stage: Start

Value or result: Left square

Evidence or stage: MOVE

Value or result: Middle square

Evidence or stage: MOVE

Value or result: Right square

Evidence or stage: STOP

Value or result: Right square

A robot starts on the left of three squares, facing right. It executes MOVE, MOVE, STOP, each MOVE being one square.

Trace the position after each command.

Middle, rightmost, unchanged. Direction, distance and starting conditions must be specified.

14. Try a new case independently

Independent case: Trace two changing number boxes

For a new display, X starts at 4 and Y starts at 1. Copy the current X into Y; add 3 to X; double Y; report both and stop. Record X and Y initially and after each instruction. Explain why Y does not become 7 when X changes. Then swap the first two instructions and trace the new procedure from the same initial state.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 9 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Instructions say pick up a pencil, write your name, then find paper. Find the first ordering problem and rewrite the three steps so another learner can follow them.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Write the first executable sorting algorithm for one easy item before adding branches or AI.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

On numbered squares 0–5, start at 1 facing increasing numbers. Execute FORWARD 2, TURN, FORWARD 1, STOP. Define TURN as reversing direction. Trace position and direction after every instruction. What if TURN were unspecified?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.